



Feature Selection in Linear Regression



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Model Selection Criteria

Having a large number of variables in a linear regression model can introduce several drawbacks, which can affect the performance, interpretability, and overall effectiveness of the model. Here are some of the key drawbacks:

- **Overfitting:** This is one of the most significant issues when including too many variables in a regression model. Overfitting occurs when the model is too complex, capturing not only the actual relationships in the data but also the noise. This leads to great performance on the training dataset but poor performance on new, unseen data.
- **Multicollinearity:** With many variables, there is a higher risk that some of them will be highly correlated with each other. Multicollinearity can make the model's estimates highly sensitive to changes in the model, and it can make it difficult to determine the effect of each variable independently. This can lead to unstable coefficients, where small changes in the data lead to large changes in the coefficients.
- **Interpretability:** A simpler model is generally easier to understand and explain. When a regression model includes many variables, it becomes more challenging to interpret the effects of each variable. This can be particularly problematic in fields where understanding the influence of variables is more important than the prediction accuracy.

Model Selection Criteria

- **Increased Variance of Estimates:** As the number of variables increases, particularly if the additional variables do not contribute much to the predictive power of the model, the variance of the estimates of the regression coefficients increases. This means that the estimates are less precise, which can decrease the statistical significance of the results.
- **Computational Complexity:** More variables mean more coefficients to estimate, which can increase the computational burden. This includes longer times for fitting models and more complex optimization problems, especially when there are constraints or when using techniques like regularization.
- **Curse of Dimensionality:** In high-dimensional spaces (many variables relative to the number of observations), various phenomena that do not occur in lower-dimensional settings become prominent. These include issues like the increased distance between data points, which can make any form of distance-based analysis less reliable.
- **Dilution of Predictive Power:** When too many irrelevant or weakly relevant variables are included in a model, they can dilute the overall predictive power by introducing more noise into the predictions.

Model Selection Criteria

There are two main ways of selecting best model from set of models

- Directly estimate prediction error using cross-validation techniques
- Indirectly estimate prediction error by making an adjustment to the training error.

k -fold Cross-validation (CV) involves randomly dividing the set of n observations into k groups (or folds) of approximately equal size.

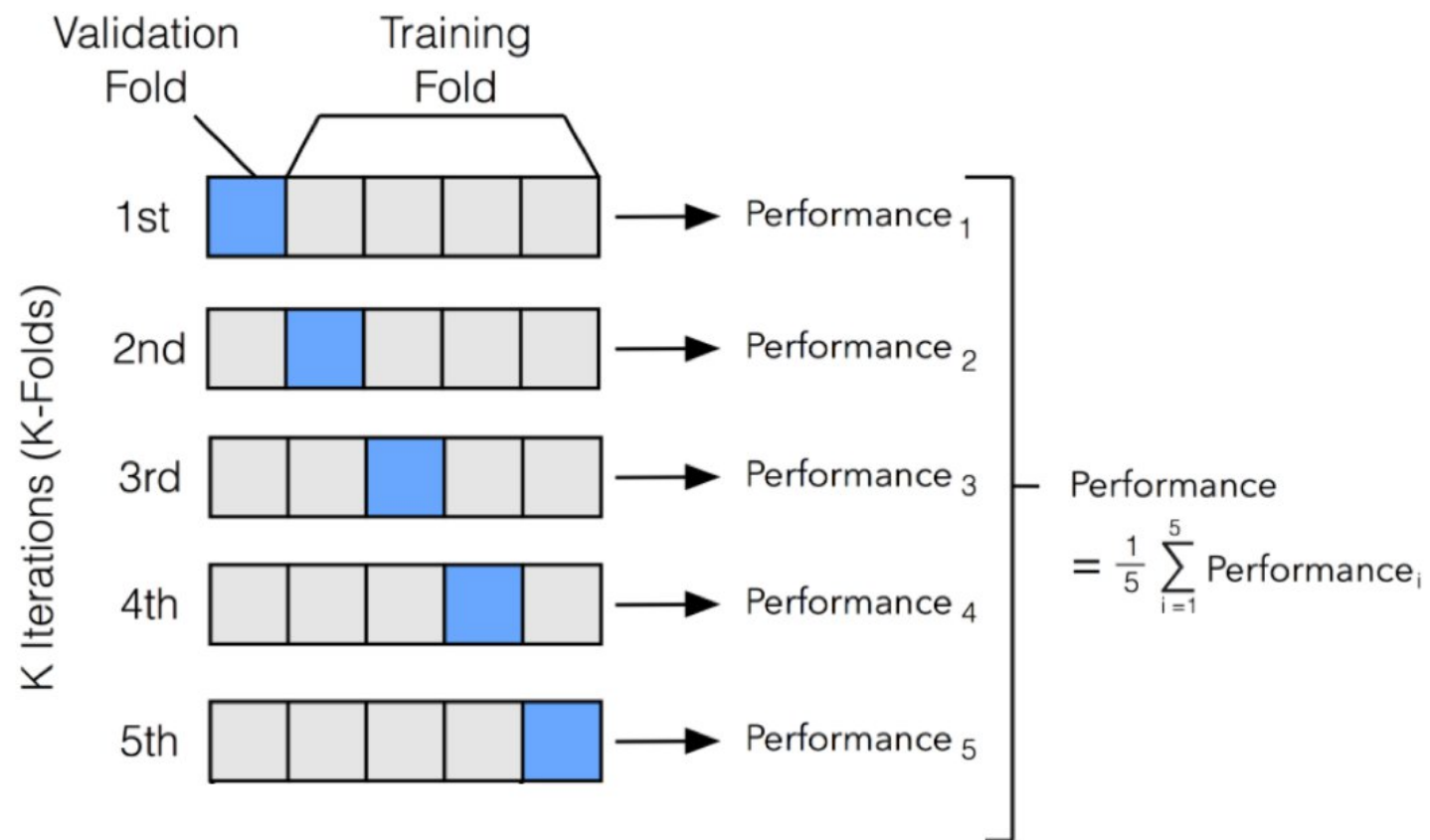
Then, for each group $i = 1, \dots, k$:

- Fold i is treated as a validation set, and the model is fit on the remaining $k - 1$ folds.
- The MSE_i is then computed based on the observations of the held out fold i .

This process results in k estimates of the test error $MSE_1, \dots, MSE_k$. The k -fold CV estimate is computed by averaging these values.

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$$

Model Selection Criteria



Model Selection Criteria

As an alternative to the direct estimation of prediction error considered in above section, we now consider a number of indirect techniques that involve adjusting the training error for the model size.

These approaches can be used to select among a set of models with different numbers of variables. We proceed to consider four such criteria.

Cp

For a given model with d predictors (out of the available p predictors) Mallows' C_p criterion is

$$C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2),$$

where RSS is the residual sum of squares for the model of interest (with d predictors), and $\hat{\sigma}^2 = RSS_p/(n - p - 1)$ is an estimate of the error variance for the full model with all p possible predictors included. As such, we use RSS_p to denote the Residual Sum of Squares for the full model with all p possible predictors included.

In practice, we choose the model which has the minimum C_p value: so we essentially penalise models of higher dimensionality (the larger d is the greater the penalty).

Model Selection Criteria

AIC

For linear models (with normal errors, as is often assumed), Mallows' C_p is equivalent to the Akaike Information Criterion (AIC) (as the two are proportional), this being given by

$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS + 2d\hat{\sigma}^2).$$

BIC

Another metric is the Bayesian information criterion (BIC)

$$BIC = \frac{1}{n\hat{\sigma}^2} (RSS + \log(n)d\hat{\sigma}^2),$$

where again the model with the minimum BIC value is selected

In comparison to C_p or AIC, where the penalty is $2d\hat{\sigma}^2$, the BIC penalty is $\log(n)d\hat{\sigma}^2$. This means that generally BIC has a heavier penalty (because $\log(n) > 2$ for $n > 7$), thus BIC selects models with fewer variables than C_p or AIC.

Model Selection Criteria

Adjusted R-Square

A simple method which is not backed up by statistical theory, but that often works well in practice, is to simply adjust the R^2 metric by taking into account the number of predictors.

The adjusted R^2 value for a model with d variables is calculated as follows

$$\text{Adjusted-}R^2 = 1 - \frac{RSS/(n - d - 1)}{TSS/(n - 1)}.$$

In this case we choose the model with the maximum adjusted R^2 value.

Best Subset Selection

To perform best subset selection, we fit a separate least squares regression for each possible combination of the p predictors. This is often broken up into stages, as follows:

1. Let M_0 denote the which contains no predictors. This model simply predicts the of the response for each observation.
2. For $k = 1, 2, \dots, p$:
 - Fit all $\binom{p}{k}$ models³ that contain exactly k predictors.
 - Pick the best among these $\binom{p}{k}$ models and call it M_k . Here, *best* is defined as having the smallest RSS or largest R^2 .
3. Select a single best model from among $M_0, M_1, \dots, M_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .

Best Subset Selection

An Illustration: Suppose we have three predictors X_1 , X_2 and X_3 . In this case best subset selection works as follows:

M_0 : intercept only (null model)

$C_1 : \begin{cases} X_1 \\ X_2 \\ X_3 \end{cases} \rightarrow$ lowest training RSS within C_1 : make this M_1

$C_2 : \begin{cases} X_1, X_2 \\ X_1, X_3 \\ X_2, X_3 \end{cases} \rightarrow$ lowest training RSS within C_2 : make this M_2

$M_3 : X_1, X_2, X_3 \sim$ (full~model)

We then choose one model among M_0, M_1, M_2, M_3 either via validation/cross-validation, or based on C_p , BIC, adjusted R^2 .

Forwards Stepwise Selection

For computational reasons, best subset selection cannot be applied with very large p as it needs to consider all 2^p models. This is true even without using cross-validation which involves fitting the coefficients of at least some of the models K times. Best subset selection may also suffer from statistical problems when p is large.

Forward stepwise selection is a computationally efficient alternative to best subset selection. This is because it considers a much smaller set of models. Forward stepwise selection works by starting with the model containing no predictors, and then adding one predictor at a time until all of the predictors are in the model. In particular, at each step the variable that gives the greatest additional improvement to the fit is added to the model. More formally, the algorithm proceeds as follows:

1. Let M_0 denote the null model which contains no predictors.
2. For $k = 0, 1, \dots, p - 1$:
 - Consider all $p - k$ models that augment the predictors in M_k with one additional predictor.
 - Choose the best among these $p - k$ models and call it M_{k+1} . Here, *best* is defined as having the smallest RSS or largest R^2 .
3. Select a single best model from among $M_0, M_1, \dots, M_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .

Forwards Stepwise Selection

An Illustration: Again, consider three predictors X_1 , X_2 and X_3 . Forward stepwise selection works as follows:

M_0 : intercept only (null model)

$C_1 : \begin{cases} X_1 \\ X_2 \\ X_3 \end{cases} \rightarrow \text{lowest training } RSS \text{ within } C_1 : \text{ make this } M_1$

$C_2 : \begin{cases} X_1, X_2 \\ X_2, X_3 \end{cases} \rightarrow \text{lowest training } RSS \text{ within } C_2 : \text{ make this } M_2$

$M_3 : X_1, X_2, X_3$ (full model)

At the end we choose again one model among M_0, M_1, M_2, M_3 based on validation/cross-validation, or based on C_p , BIC, adjusted R^2 .

Forwards Stepwise Selection

- Best subset selection requires training 2^p models, forward stepwise selection requires only $1 + \frac{p(p+1)}{2}$ comparisons.
- So for $p = 20$: best subset selection considers 1,048,576 models, whereas forward stepwise selection considers 211 models!
- There is a cost to pay though for the gain in scalability. Forward stepwise selection can be sub-optimal in the sense that the models selected are not the best subset models.
- This is because the guided forward stepwise search at each step depends on the previously selected predictor.
- As an example, in the illustration above we had $p = 3$. The best possible one-variable model contained X_2 . It is quite possible that the best two-variable model contains the pair (X_1, X_3) , however, we did not consider this combination in the above algorithm.
- However, in practice best subset selection and forward stepwise selection perform quite similarly in terms of predictive accuracy.

Backwards Stepwise Selection

Backwards stepwise selection begins with the full model containing all p predictors, and iteratively removes the least useful predictor one-at-a-time.

1. Let M_p denote the which contains all predictors.
2. For $k = p, p - 1, \dots, 1$:
 - Consider all k models that contain all but one of the predictors in M_k , for a total of $k - 1$ predictors.
 - Choose the best among these k models and call it M_{k-1} . Here, *best* is defined as having the smallest RSS or largest R^2 .
3. Select a single best model from among $M_0, M_1, \dots, M_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .

Just like the forward algorithm, backward elimination requires only $1 + \frac{p(p+1)}{2}$ comparisons.

Backwards Stepwise Selection

An Illustration: Again, consider three predictors X_1 , X_2 and X_3 . Backwards stepwise selection works as follows:

$M_3 : X_1, X_2, X_3$ (full model)

$C_2 : \begin{cases} X_1, X_2 \rightarrow \text{lowest training } RSS \text{ within } C_2 : \text{ make this } M_2 \\ X_1, X_3 \\ X_2, X_3 \end{cases}$

$C_1 : \begin{cases} X_1 \\ X_2 \rightarrow \text{lowest training } RSS \text{ within } C_1 : \text{ make this } M_1 \end{cases}$

M_0 : intercept only (null model)

At the end we choose again one model among M_0, M_1, M_2, M_3 based on cross-validation, or based on C_p , BIC, adjusted R^2 .